

Mental Health Analysis and Chatbot

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I. ABSTRACT

Mental health conditions affect a growing share of the global population, yet timely diagnosis remains a challenge in most healthcare settings. This project explores whether modern natural language processing (NLP) techniques can help bridge that gap by automatically classifying social-media text into seven mental health categories: Normal, Anxiety, Depression, Stress, Suicidal ideation, Bipolar disorder, and Personality disorder. We train and compare four distinct model configurations on a large-scale, Reddit-sourced dataset of over 53,000 annotated posts: (i) a domain-adapted MentalRoBERTa model with weighted cross-entropy loss, (ii) the same backbone paired with Focal Loss [1], (iii) RoBERTa-Large fine-tuned through Low-Rank Adaptation (LoRA) [2], and (iv) a novel hybrid architecture that combines RoBERTa embeddings with a Bidirectional LSTM and multi-kernel CNN [3]. Among these, the hybrid model delivers the strongest results—reaching 85.52% accuracy and a macro F1-score of 0.841—demonstrating that extracting features from multiple perspectives simultaneously yields richer text representations than relying on a single classification head. The findings offer a practical framework for building NLP-driven mental health screening tools that can operate under real-world data constraints [4][5].

Index Terms—Mental Health, Natural Language Processing, RoBERTa, Deep Learning, Text Classification, Transformer, BiLSTM, CNN, Sentiment Analysis, Class Imbalance

II. INTRODUCTION

Mental health disorders are among the most widespread yet under-addressed public health challenges of our time. The World Health Organization estimates that roughly one in every eight people worldwide lives with a mental health condition, and the actual numbers are almost certainly higher because many individuals never seek professional help [6]. The consequences extend well beyond personal suffering: untreated mental illness contributes to reduced workplace productivity, strained healthcare systems, and elevated mortality—particularly through its link to suicidal ideation and self-harm [7].

The challenge is compounded by the fact that traditional diagnostic pathways are inherently bottlenecked. A typical mental health assessment relies on a patient seeking help, securing an appointment with a trained professional, and honestly disclosing their symptoms—each of these steps introduces friction and potential for dropout. In many parts of the world, there are simply not enough trained clinicians to meet the demand. Computational methods that can process naturalistic language data at scale offer a way to complement these traditional pipelines, potentially catching signs of distress earlier and more broadly than clinical encounters alone [9][13].

In recent years, digital communication platforms—especially social media—have opened up an entirely new avenue for understanding mental health at scale. Every day, millions of people share their thoughts, emotions, and personal experiences on platforms like Reddit and Twitter. This enormous volume of naturalistic text carries subtle linguistic markers of psychological distress that are often invisible in formal clinical encounters [8]. If we can learn to detect those markers automatically, we gain a powerful complement to traditional clinical assessments, which are limited by clinician availability, self-reporting biases, and the simple fact that most people do not see a therapist regularly.

Data mining and machine learning provide the tools needed to extract meaningful patterns from this kind of unstructured textual data. Early approaches relied on bag-of-words features and hand-crafted rules, but these methods struggled to capture the nuanced, context-dependent nature of mental health language [9]. The landscape changed dramatically with the arrival of transformer-based language models such as BERT and RoBERTa, which learn rich contextual representations of text and have set new performance benchmarks across a wide range of NLP tasks [10]. More recently, researchers have pushed performance even further by using domain-adaptive pretraining, where a general-purpose language model is given additional training on domain-specific text before being fine-tuned on the actual task [11].

That said, applying these models to multi-class mental health classification is not straightforward. Real-world mental health datasets tend to be severely imbalanced—conditions like depression dominate the data, while rarer but clinically significant categories such as personality disorder are underrepresented [12]. The linguistic boundaries between related conditions (depression vs. anxiety, stress vs. suicidal ideation) are often blurry, making it hard for a model to learn clean decision boundaries. And from a practical standpoint, large transformer models can be expensive to fine-tune, calling for parameter-efficient strategies that do not sacrifice too much accuracy.

This project tackles these challenges head-on. We systematically compare four model configurations, each embodying a different adaptation strategy: domain-adapted fine-tuning, cost-sensitive loss modifications, parameter-efficient adaptation via LoRA, and a hybrid architecture that combines global, sequential, and local feature extraction into a single pipeline. By benchmarking all four approaches on the same dataset and evaluation protocol, we aim to identify which design choices matter most for mental health text classification and to provide practical guidance for anyone looking to deploy NLP-based screening tools in this domain [13].

III. PROBLEM STATEMENT

At its core, the goal of this project is straightforward: given a social-media post written by an anonymous user, classify it into one of seven mental health categories—Normal, Anxiety, Depression, Stress, Suicidal ideation, Bipolar disorder, or Personality disorder. This is a supervised multi-class classification problem, and while the formulation is simple, the practical challenges are anything but.

The first major challenge is **class imbalance**. In our dataset, the "Normal" and "Depression" categories together account for over 60% of all samples, while "Personality disorder" makes up less than 2%. A naive classifier could achieve decent overall accuracy simply by predicting the majority class for everything, which would be useless for the minority categories that often carry the greatest clinical urgency [12]. The second challenge is **semantic overlap**. Mental health conditions frequently co-occur and share overlapping linguistic expressions. Someone describing suicidal thoughts will almost certainly use depressive language, and posts about anxiety and stress can look remarkably similar at the surface level [14]. The classifier needs to pick up on subtle differences in phrasing, context, and tone to distinguish between these categories. The third challenge is **computational cost**. Modern transformer models like RoBERTa-Large contain hundreds of millions of parameters, and fully fine-tuning them requires significant GPU resources. Parameter-efficient methods like LoRA are attractive for reducing this cost, but the trade-off between efficiency and accuracy is not always clear [2].

To address these challenges, we evaluate four complementary strategies: domain-adaptive pretraining for better feature quality, cost-sensitive loss functions to handle class imbalance, parameter-efficient fine-tuning to manage computational requirements, and a hybrid architecture that combines multiple feature extraction paradigms. The overarching goal is to find the most effective and practically deployable approach for automated mental health screening from social media text.

IV. RELATED WORK

Our work builds directly on several key lines of research in NLP, mental health computing, and cost-sensitive learning. This section reviews the four papers that most directly influenced our model design, followed by a broader survey of the relevant literature.

A. *MIRoBERTa: Mental Illness Text Classification With Transfer Learning on Subreddits*

The work by Sao and Lim [1] is arguably the closest predecessor to our project. They introduced MIRoBERTa, a domain-adapted version of RoBERTa trained specifically on mental health Reddit posts, and used it to classify text into eight mental illness categories (ADHD, Anxiety, Bipolar, BPD, Depression, OCD, PTSD, and None). What makes their work especially relevant is the scale and rigor of their experimental pipeline: they compared traditional ML baselines (Logistic Regression, Naive Bayes, Linear SVM), deep learning models (BiLSTM, BiGRU with GloVe), and multiple transformers (BERT, RoBERTa, BigBird, Longformer) on a dataset of 54,161 Reddit submissions.

Their key finding was that domain-adaptive pretraining—continuing the masked language modeling objective on a 243,335-post mental health corpus—yielded the best individual model (0.847 F1), outperforming all general-purpose alternatives. An ensemble of the original and domain-adapted RoBERTa pushed performance further to 0.851 F1. They also used SHAP analysis to identify disorder-specific linguistic markers, adding an interpretability dimension to their work. This paper validated the domain-specific pretraining strategy that we adopt through the MentalRoBERTa checkpoint, and our project extends their direction by exploring hybrid architectures, alternative loss functions, and parameter-efficient fine-tuning.

B. *Improving Sentiment Classification Using a RoBERTa-Based Hybrid Model*

Semary et al. [3] proposed a hybrid architecture that combines RoBERTa embeddings with a CNN-LSTM pipeline for sentiment classification. The basic idea is compelling: RoBERTa produces high-quality contextual embeddings, but a single classification head on top of the CLS token may not capture all the useful information in those embeddings. By routing the embeddings through both a CNN (for local n-gram patterns) and an LSTM (for sequential dependencies), the model can extract richer, multi-perspective features. They evaluated this approach on the Twitter US Airlines and IMDb datasets, achieving 96.28% accuracy on IMDb and 94.2% on Twitter, comfortably beating standalone RoBERTa, CNN, and LSTM baselines.

This paper served as the direct inspiration for our hybrid Model 4. We adapted the core architecture with several modifications: a bidirectional LSTM instead of a unidirectional one, multi-kernel CNN filters to capture patterns at different granularities, and a three-branch concatenation strategy that also preserves the original CLS token embedding alongside the sequential and convolutional feature streams.

C. *MentalBERT: Publicly Available Pretrained Language Models for Mental Healthcare*

Ji et al. [11] made a significant contribution to the field by releasing MentalBERT and MentalRoBERTa—BERT and RoBERTa models that had been further pretrained on a large corpus of Reddit mental health posts. They evaluated these checkpoints on several downstream benchmarks, including depression detection, stress detection, and suicidal ideation classification, and found that domain-specific pretraining consistently improved performance over general-purpose models. We directly use the MentalRoBERTa checkpoint from their work as the backbone for our Models 1 and 2, which saves us the significant computational cost of performing domain-adaptive pretraining from scratch.

D. *A Survey of Methods for Addressing Class Imbalance in Deep-Learning Based NLP*

Henning et al. [12] provided a comprehensive survey of techniques for dealing with class imbalance in NLP. They organized the landscape into sampling methods, data augmentation, loss function modifications (weighted cross-entropy, Focal Loss), staged learning, and model design choices. One of their key findings—and one that directly shaped our approach—is that loss-level modifications like weighted cross-entropy and Focal Loss are both effective and low-overhead: they require minimal changes to the training pipeline while providing meaningful improvements in minority-class performance [15]. This motivated our decision to focus on loss function strategies rather than more complex sampling or augmentation approaches.

E. Additional Related Literature

Several other works informed our understanding of this space. Singh et al. [5] offered a foundational overview of sentiment analysis and emotion detection techniques. Onyenwe et al. [6] examined pre-processing strategies for noisy mental health text. Hagerova et al. [7] demonstrated that even informal writing samples can reveal underlying depression. Zucco et al. [8] explored the challenges of using large-scale social media data for community-level mental health prediction. On the machine learning side, Hasan et al. [9] presented a systematic review comparing ML algorithms for mental health detection, Febrian and Awangga [16] highlighted the appeal of Random Forest and Logistic Regression for interpretable clinical predictions, and Chen et al. [17] analyzed which demographic and psychological factors best predict distress. Bielinski et al. [14] provided useful benchmarking data for algorithm comparison. For deep learning, an IEEE Conference paper [18] reported up to 97% accuracy with hybrid CNN-LSTM models for depression detection, Lekkas et al. [13] showed how mobile apps can track mental health trends among university students, and Maekawa et al. [19] discussed Bayesian screening tools for depression.

V. PRELIMINARY CONCEPTS

Before diving into the methodology, it helps to establish the theoretical groundwork for the techniques used in this project. This section walks through the key concepts, from the base language model to the specific architectural choices and training strategies.

A. RoBERTa (Robustly Optimized BERT Pretraining Approach)

RoBERTa, developed by Meta AI in 2019 [10], is an optimized version of BERT that learns contextual text representations through Masked Language Modeling (MLM)—a self-supervised objective where random tokens in the input are replaced with a mask, and the model learns to predict the original tokens from context. Both BERT and RoBERTa are built on the Transformer encoder architecture, which uses multi-head self-attention to capture relationships between tokens regardless of their distance in the input.

Where RoBERTa differs from BERT is in the details of its training procedure. It uses dynamic masking (a fresh random mask for each training pass, rather than BERT’s static masks), drops the Next Sentence Prediction objective (which turned out to be unhelpful), and trains on substantially more data with larger batch sizes and for longer. These seemingly incremental changes produced a meaningful jump in performance, allowing RoBERTa to achieve state-of-the-art results on the GLUE benchmark and making it a go-to backbone for text classification [10]. In this project, we use RoBERTa-base (125M parameters, 12 layers, 768-dimensional hidden states) and RoBERTa-Large (355M parameters, 24 layers, 1024-dimensional hidden states), along with the domain-adapted MentalRoBERTa variant.

B. Domain-Adaptive Pretraining

General-purpose models like RoBERTa are pretrained on broad corpora—news articles, books, Wikipedia—that do not necessarily reflect the vocabulary and writing patterns found in mental health discourse. Domain-adaptive pretraining addresses this gap by continuing the MLM training on domain-specific text before the model is fine-tuned on the downstream task. For mental health NLP, this means training on large collections of Reddit posts from subreddits like r/depression, r/anxiety, and r/SuicideWatch [11]. The result is a model that has already absorbed disorder-specific terminology, common expressions of distress, and contextual cues that general-purpose models might miss. Both MentalRoBERTa [11] and MIRoBERTa [1] were produced this way, and both consistently outperform their general-purpose counterparts.

C. Weighted Cross-Entropy Loss

Standard cross-entropy loss does not distinguish between classes—a misclassification of a rare class contributes the same to the total loss as a misclassification of a common one. When the dataset is heavily imbalanced (as ours is), this causes the model to optimize mainly for majority classes. Weighted cross-entropy fixes this by multiplying each class’s loss by a weight that is inversely proportional to its frequency. In practice, if class c appears f_c times in a dataset of N total samples across C classes, its weight is $w_c = N / (C \times f_c)$. This means a mistake on a rare class like Personality disorder generates a much larger gradient signal than a mistake on the abundant Normal class, pushing the model to pay more attention to the minorities [12].

D. Focal Loss

Focal Loss, introduced by Lin et al. [15] in the context of object detection, takes the idea of cost-sensitive learning a step further. Instead of just reweighting by class frequency, it also down-weights “easy” samples—those that the model is already confident about—and focuses training on the harder, more uncertain examples. The formula is $FL(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t)$, where p_t is the predicted probability of the correct class, α handles class-level reweighting, and γ (the focusing parameter) controls how aggressively easy samples are down-weighted. With $\gamma = 0$, Focal Loss reduces to standard cross-entropy; the original paper found $\gamma = 2.0$ to work well for most tasks. In our Model 2, we use $\gamma = 2.0$ and set α to the inverse-frequency class weights [4][15].

E. Low-Rank Adaptation (LoRA)

Fine-tuning a model with hundreds of millions of parameters is not always feasible, especially on consumer-grade GPUs. LoRA [2] offers an elegant solution: instead of updating all the weights, it freezes the pretrained model and injects small, trainable low-rank matrices into specific layers. For a weight matrix W of size $d \times k$, LoRA decomposes the update as $\Delta W = B \times A$, where B is $d \times r$ and A is $r \times k$, with r being much smaller than d or k . Only B and A are trained, which can reduce the number of trainable parameters by orders of magnitude—from hundreds of millions down to less than a million. At inference time, these matrices are merged back into the original weights, so there is no speed penalty. In our Model 3, we apply LoRA to the query and value projection matrices in each attention layer of RoBERTa-Large with rank $r = 16$.

F. Bidirectional LSTM (BiLSTM)

LSTM networks are a type of recurrent neural network specifically designed to handle the vanishing gradient problem that plagues standard RNNs. They use gating mechanisms—an input gate, a forget gate, and an output gate—to control what information is retained or discarded at each time step, making them effective at capturing long-range dependencies in sequential data. A Bidirectional LSTM processes the sequence in both directions (left-to-right and right-to-left), concatenating the two hidden states at each position. This gives the model access to both past and future context, which is especially useful in text classification where a word’s meaning often depends on what comes both before and after it [3]. In our hybrid model, the BiLSTM processes the full sequence of RoBERTa token embeddings to extract sequential patterns that complement the transformer’s self-attention.

G. 1D Convolutional Neural Network (CNN) for Text

One-dimensional CNNs slide convolutional filters across a sequence of token embeddings to detect local patterns—essentially n-grams at the feature level. A filter of kernel size k looks at k consecutive tokens at a time, learning to recognize patterns like negation phrases, intensifiers, or diagnostic keyword combinations. By using multiple kernel sizes (e.g., 2, 3, 4, 5), the network detects patterns at several different granularities simultaneously. After convolution, a ReLU activation introduces nonlinearity, and max-over-time pooling picks the strongest activation from each filter across the entire sequence, producing a fixed-length feature vector regardless of input length [3]. In our hybrid model, this CNN branch captures local discriminative patterns that might be missed by the global pooling of the transformer or the sequential processing of the BiLSTM.

H. Early Stopping and Model Selection

Training deep neural networks for too many epochs almost always leads to overfitting—the model starts memorizing training data instead of learning generalizable patterns. Early stopping addresses this by monitoring a validation metric during training and halting when performance stops improving for a set number of evaluation rounds (the patience). In our experiments, we use macro F1 as the stopping criterion because, unlike accuracy, it treats all classes equally regardless of frequency—which matters a great deal when you have a class that is 30% of the data sitting next to one that is less than 2% [12].

VI. METHODOLOGY AND IMPLEMENTATION

A. Dataset Description

The dataset for this study is the “Sentiment Analysis for Mental Health” collection published on Kaggle by Suchintika Sarkar. It brings together over 53,000 text statements sourced from Reddit and Twitter, each annotated into one of seven mental health categories: Normal, Anxiety, Depression, Stress, Suicidal, Bipolar, and Personality disorder [8]. We chose this dataset primarily for its scale (large enough to fine-tune data-hungry transformer models), its seven-class taxonomy (broad enough to be clinically meaningful), and its inherent class imbalance (realistic enough to test cost-sensitive learning strategies).

After preprocessing and stratified splitting, we end up with a training set of 35,058 samples and a validation set of 7,512 samples. Table I breaks down the class distribution, which clearly illustrates the imbalance problem: Normal and Depression together account for roughly 60% of the data, while Personality disorder represents less than 2%.

TABLE I. Dataset Class Distribution

Category	Label	Train	Val	Total	%
Normal	0	10,716	2,296	13,012	30.6
Depression	2	10,462	2,241	12,703	29.8
Suicidal	4	7,397	1,585	8,982	21.1
Anxiety	1	2,515	539	3,054	7.2
Bipolar	5	1,744	374	2,118	5.0
Stress	3	1,602	343	1,945	4.6
Personality Disorder	6	622	134	756	1.8
Total	—	35,058	7,512	42,570	100.0

B. Data Preprocessing

The raw social-media text goes through several cleaning steps before it reaches the model. We strip out URLs, email addresses, and excess whitespace; decode HTML entities; and remove user mentions and hashtag symbols while keeping the underlying text (since those words can carry sentiment-relevant information) [5]. Tokenization is handled by the RoBERTa tokenizer, which uses Byte-Pair Encoding to convert text into subword tokens. All sequences are truncated to a maximum of 512 tokens—the positional embedding limit of RoBERTa—and we use dynamic padding so each batch is padded only to the length of its longest sequence, avoiding wasted computation on shorter posts [10].

To deal with class imbalance, we compute inverse-frequency class weights from the training set: for a class c with frequency f_c in a dataset of N total samples and C classes, the weight is $w_c = N / (C \times f_c)$. These weights get applied either as coefficients in the weighted cross-entropy loss or as the α parameter in Focal Loss, ensuring that mistakes on rare classes generate proportionally larger gradient signals during training [12][15].

C. Model Architectures

We evaluate four model configurations, each representing a fundamentally different approach to adapting pretrained language models for this task. All four share the same dataset, preprocessing, and evaluation protocol, making the comparison as fair as possible. Table II summarizes the configurations, and the architectures are described in detail below.

TABLE II. Summary of Model Configurations

Model	Backbone	Key Strategy	Loss Function	Trainable Params
1	MentalRoBERTa-base	Domain adaptation	Weighted CE	~125M
2	MentalRoBERTa-base	Focal Loss	Focal Loss + α	~125M
3	RoBERTa-Large + LoRA	Parameter-efficient	Weighted CE	~0.8M
4	RoBERTa-base + BiLSTM + CNN	Hybrid features	Weighted CE	~130M

1) Model 1: MentalRoBERTa with Weighted Cross-Entropy

Our baseline uses MentalRoBERTa (mental/mental-roberta-base), a RoBERTa model that has been further pretrained on mental health subreddit text [11]. On top of the transformer encoder, we place the standard classification head: a dropout layer followed by a linear projection from the 768-dimensional hidden space to seven output classes. The loss function is weighted cross-entropy with inverse-frequency class weights. This configuration isolates the contribution of domain-adaptive pretraining—everything else is as vanilla as possible, giving us a clean baseline to compare against.

2) Model 2: MentalRoBERTa with Focal Loss

Model 2 is architecturally identical to Model 1—same backbone, same classification head—but swaps the loss function for Focal Loss with $\gamma = 2.0$ and class weights as the α parameter [15]. The idea is to test whether a more sophisticated cost-sensitive strategy can do better than simple inverse-frequency weighting. By dynamically down-weighting samples that the model is already confident about, Focal Loss should, in theory, direct more training effort toward the harder, more ambiguous examples that often come from minority classes.

3) Model 3: RoBERTa-Large with LoRA

This configuration asks a different question: can a larger pretrained backbone compensate for constrained fine-tuning? We use RoBERTa-Large (355M parameters) but freeze all the original weights and apply LoRA [2] only to the query and value attention projections. With a rank of $r = 16$, only about 0.8M parameters are actually trainable, making the setup feasible on a single consumer GPU. We had to reduce the maximum sequence length and batch size to fit within memory constraints, which is an important caveat—longer posts get truncated more aggressively than in the other configurations.

4) Model 4: RoBERTa-Base + BiLSTM-CNN Hybrid

The hybrid model is the centerpiece of this project, inspired directly by Semary et al. [3]. Instead of placing a single classification head on top of RoBERTa’s output, we extract three complementary feature streams from the encoder’s token embeddings. The first stream takes the CLS token embedding as a global sentence-level representation. The second passes the full token sequence through a Bidirectional LSTM with mask-weighted mean pooling and layer normalization, capturing sequential dependencies. The third applies 1D convolutions with kernel sizes of 2, 3, 4, and 5 to extract local n-gram patterns, followed by ReLU activation and max-over-time pooling. These three streams are concatenated, passed through layer normalization and dropout ($p = 0.3$), and projected to seven output classes via a linear layer.

D. Hybrid Model Methodology and Training Configuration

The hybrid model’s training pipeline proceeds through six stages. Stage 1 (Data Preparation) loads the preprocessed CSV files, label mappings, and class weight configurations. Stage 2 (Tokenization) applies the RoBERTa-base tokenizer with truncation to 512 tokens; dynamic padding via a data collator pads each batch to its longest sequence. Stage 3 (Model Construction) initializes the pretrained RoBERTa-base encoder, BiLSTM layer, CNN filter bank, normalization layers, dropout, and the linear classifier. Stage 4 (Training) uses a custom Trainer that applies weighted cross-entropy loss with mixed-precision (FP16) computation, gradient clipping, cosine learning rate scheduling, and early stopping based on macro F1. Stage 5 (Evaluation) selects the best checkpoint by macro F1 and computes accuracy, macro/weighted precision, recall, F1, and per-class F1 scores. Stage 6 (Logging) saves training curves and all metrics to JSON and image files for analysis.

TABLE III. Hyperparameters Configuration for the Hybrid Model (Model 4)

Hyperparameter	Value
Base Model	FacebookAI/roberta-base
Maximum Sequence Length	512 tokens
Number of Labels	7
Batch Size	8
Number of Epochs	4
Learning Rate	2e-5
Weight Decay	0.01
Warmup Ratio	0.1
Gradient Accumulation Steps	4
Evaluation Steps	200
Logging Steps	100
Save Total Limit	2
Early Stopping Patience	4

All four models use the AdamW optimizer with these consistent settings, with minor adjustments for Models 2 and 3 to accommodate their specific requirements. Model 2 additionally requires the γ focusing parameter (set to 2.0) for Focal Loss, and Model 3 requires LoRA-specific parameters including the rank ($r = 16$), target modules (query and value projections), and a reduced sequence length to fit within GPU memory constraints.

Fig. 1 presents the overall experimental workflow of this project, from the raw dataset through preprocessing, parallel model training, validation evaluation, and final model selection.

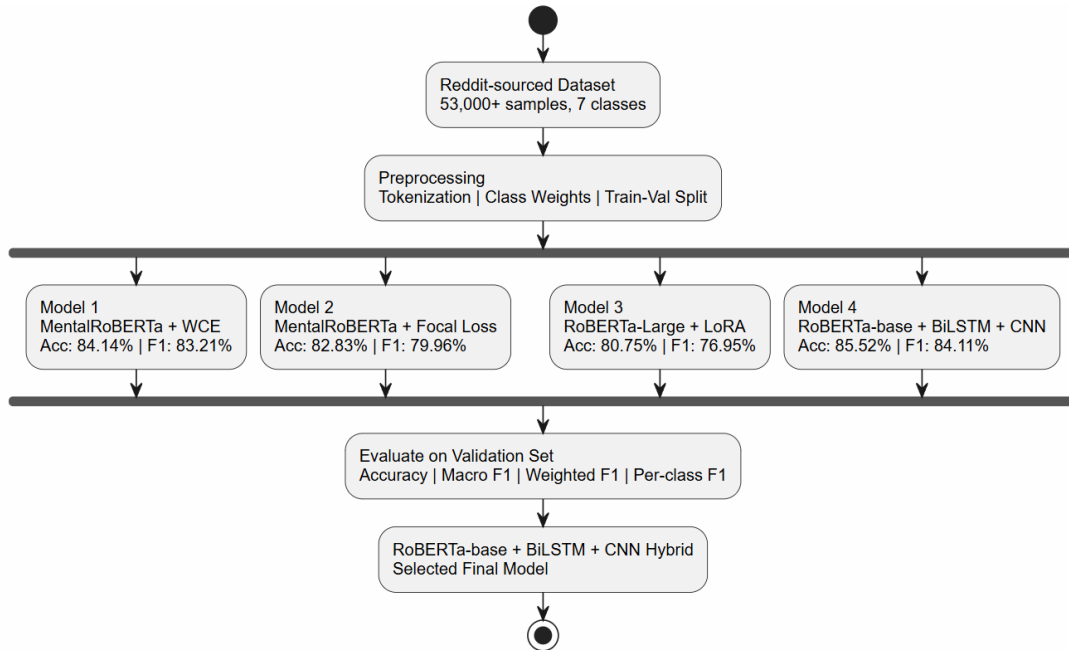


Fig. 1. Experimental Workflow: Dataset to Final Model Selection

Fig. 2 shows the conceptual architecture of the hybrid model: input text flows through the RoBERTa tokenizer and encoder, producing token embeddings that feed into three parallel branches (CLS, BiLSTM, CNN), whose outputs are concatenated and projected to the final classification.

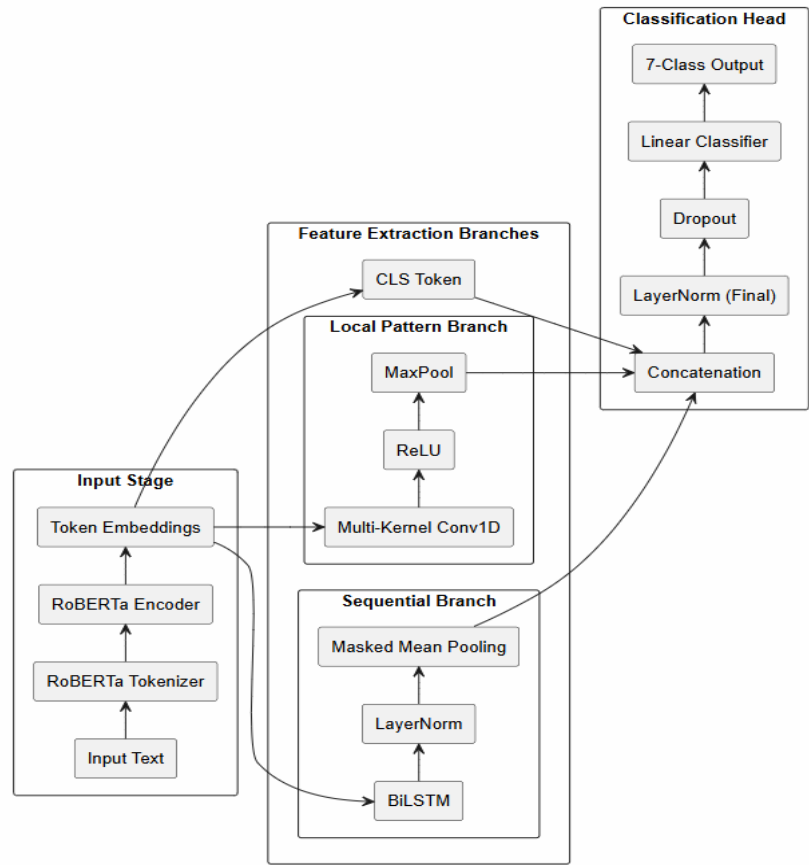


Fig. 2. System Architecture of the Hybrid RoBERTa-BiLSTM-CNN Model (Model 4)

E. Deployed Application: Mental Health Chatbot

To demonstrate the practical utility of the trained hybrid model, we developed a real-time Mental Health Assistant chatbot built with Streamlit. The application integrates two complementary intelligence systems: the locally deployed RoBERTa-BiLSTM-CNN hybrid classifier for real-time text classification, and the Gemini 2.5 Flash API for generating empathetic, context-aware conversational responses. Fig. 3 illustrates the system architecture of this chatbot application.

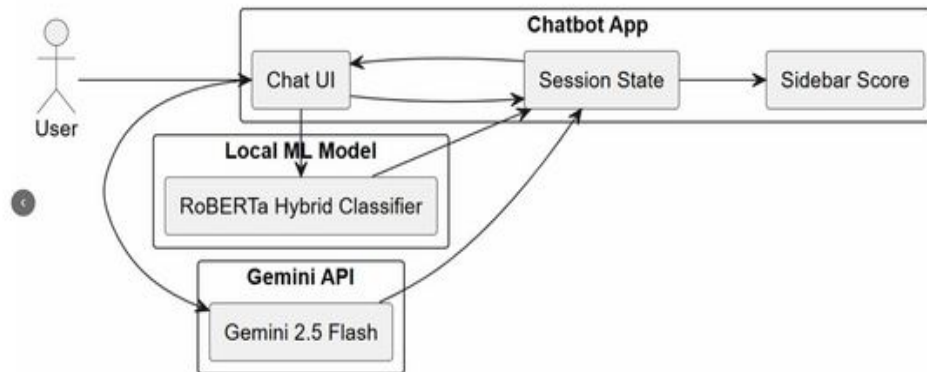


Fig. 3. System Architecture of the Mental Health Chatbot Application

When a user sends a message, it is simultaneously processed by the local ML model (which produces a classification label, confidence score, and per-class probabilities) and by the Gemini API (which generates a supportive reply and an independent label assessment). The two risk signals are blended using a weighted formula: $combined_{risk} = 0.65 \times ML_{risk} + 0.35 \times LLM_{\hat{c}}$. This combined risk is accumulated into a session-level Mental Health Score that tracks the user’s emotional trajectory over the course of the conversation.

The application operates in four distinct modes based on the current risk level: **normal** (score ≤ 0.4 , light and friendly tone), **elevated** (score 0.4–0.7, warm and validating), **high_risk** (score > 0.7 , empathetic with professional help suggestions), and **crisis** (suicidal probability > 0.5 , immediate safety resources). The sidebar displays real-time model predictions, confidence scores, suicidal probability, and the cumulative mental health score with appropriate clinical guidance. A disclaimer prominently reminds users that the system is not a substitute for professional care.

VII. MODEL COMPARISON ANALYSIS

With all four models trained under the same conditions, we can now compare their performance on the held-out validation set. Table III presents the overall metrics, and Table IV gives a per-class breakdown of F1-scores.

TABLE IV. Final Validation Metrics for All Models

Model	Accuracy	P (macro)	R (macro)	F1 (macro)	F1 (wtd)	Ref.
Model 1: MentalRoBERTa + WCE	0.8415	0.8369	0.8412	0.8321	0.8408	[1][11]
Model 2: MentalRoBERTa + FL	0.8283	0.7789	0.8324	0.7996	0.8264	[1][15]
Model 3: RoBERTa-Large + LoRA	0.8075	0.7657	0.7761	0.7695	0.8076	[2]
Model 4: Hybrid (Best)	0.8552	0.8317	0.8540	0.8411	0.8557	[3]

TABLE V. Per-Class F1-Scores for All Models

Model	Normal	Anxiety	Depress.	Stress	Suicidal	Bipolar	PD
Model 1	0.962	0.897	0.756	0.814	0.771	0.879	0.746
Model 2	0.948	0.878	0.753	0.780	0.762	0.819	0.656
Model 3	0.927	0.817	0.762	0.686	0.734	0.816	0.644
Model 4	0.958	0.900	0.802	0.801	0.780	0.892	0.755

The headline result is clear: the hybrid model (Model 4) delivers the best overall performance, with 85.52% accuracy and a macro F1 of 0.841. This confirms our hypothesis that combining multiple feature extraction perspectives—global self-attention, sequential context, and local n-gram patterns—produces richer representations than a single classification head [3]. The domain-adapted MentalRoBERTa baseline (Model 1) is a respectable second at 84.15% accuracy, which underscores just how much domain-specific pretraining contributes [11].

Focal Loss (Model 2) actually hurts performance compared to standard weighted cross-entropy, dropping accuracy to 82.83% and macro F1 to 0.800. This suggests that for this particular dataset and level of class imbalance, the inverse-frequency weighting already provides enough minority-class emphasis, and the additional focusing mechanism may introduce optimization instability [12][15]. LoRA-adapted RoBERTa-Large (Model 3) comes in last at 80.75% accuracy, suggesting that parameter-efficient fine-tuning with only ~0.8M trainable parameters does not give the model enough room to adapt, despite the larger backbone’s richer representations [2].

Looking at the per-class F1 breakdown in Table IV, the hybrid model’s gains are not limited to one or two categories—it improves on virtually every class. The most striking improvements are in Depression (0.802 vs. 0.756 for the baseline), Bipolar (0.892 vs. 0.879), and Personality disorder (0.755 vs. 0.746). The fact that even the rarest class benefits from the hybrid architecture speaks to the value of multi-perspective feature extraction for challenging minority categories [14].

VIII. RESULTS AND OUTPUT

This section presents the training dynamics and performance visualizations for each model. All figures show four panels: training/validation loss curves, accuracy progression, macro F1 over training steps, and the final per-class F1 distribution.

A. Model 1: MentalRoBERTa with Weighted Cross-Entropy

Model 1 converges steadily, with validation loss dropping through the early epochs before plateauing around step 1500. Accuracy climbs quickly to about 84% and holds. The per-class F1 bars show exactly what we would expect: the Normal class dominates (0.962), Anxiety is strong (0.897), and the minority classes—Depression (0.756) and Personality disorder (0.746)—are harder but still respectable. The macro F1 stabilizes around 0.832, making this a solid baseline [1][11].

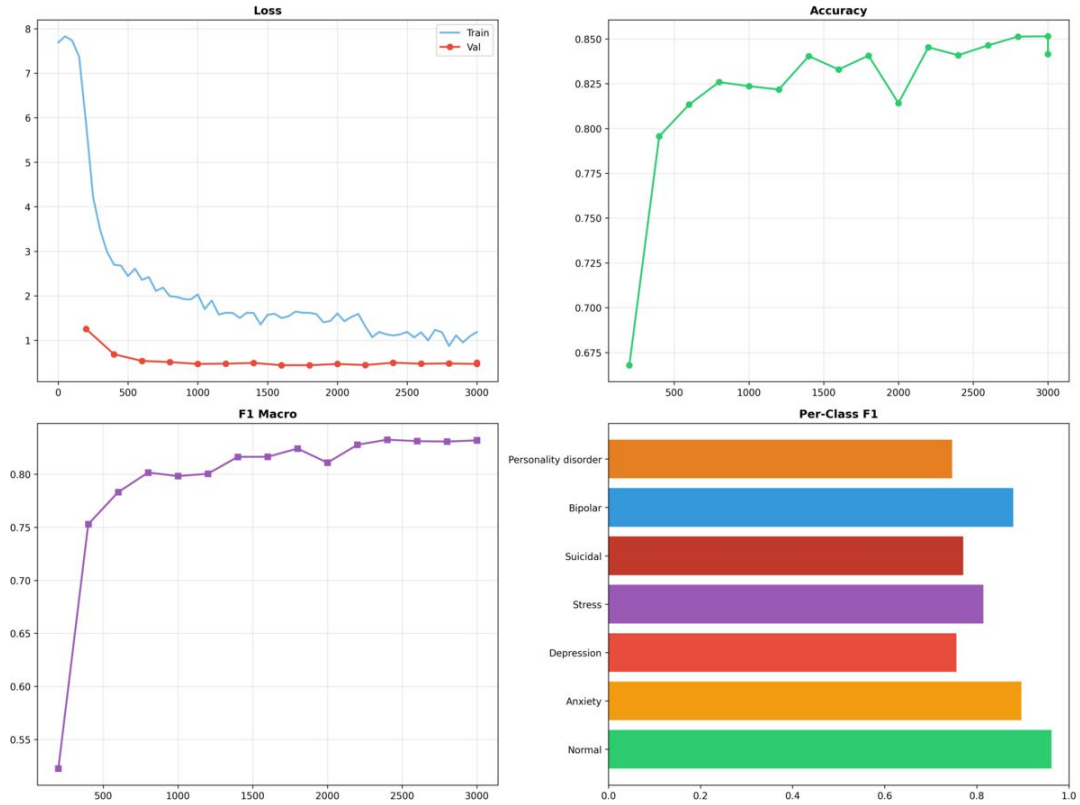


Fig. 4. Model 1 Combined Training Curves: Loss, Accuracy, F1 Macro, and Per-Class F1

B. Model 2: MentalRoBERTa with Focal Loss

Focal Loss produces a noticeably different training profile. The loss curve shows a more gradual initial decrease—consistent with the focusing mechanism suppressing easy-sample gradients early on—followed by continued refinement. Accuracy tops out at roughly 83%, and while Focal Loss does achieve competitive recall on some minority classes (suggesting it is indeed focusing on harder examples), this comes at the cost of precision, pulling the overall F1 down to 0.800. The Bipolar class drops to 0.819 and Personality disorder falls to 0.656, a substantial decrease from the baseline [15].

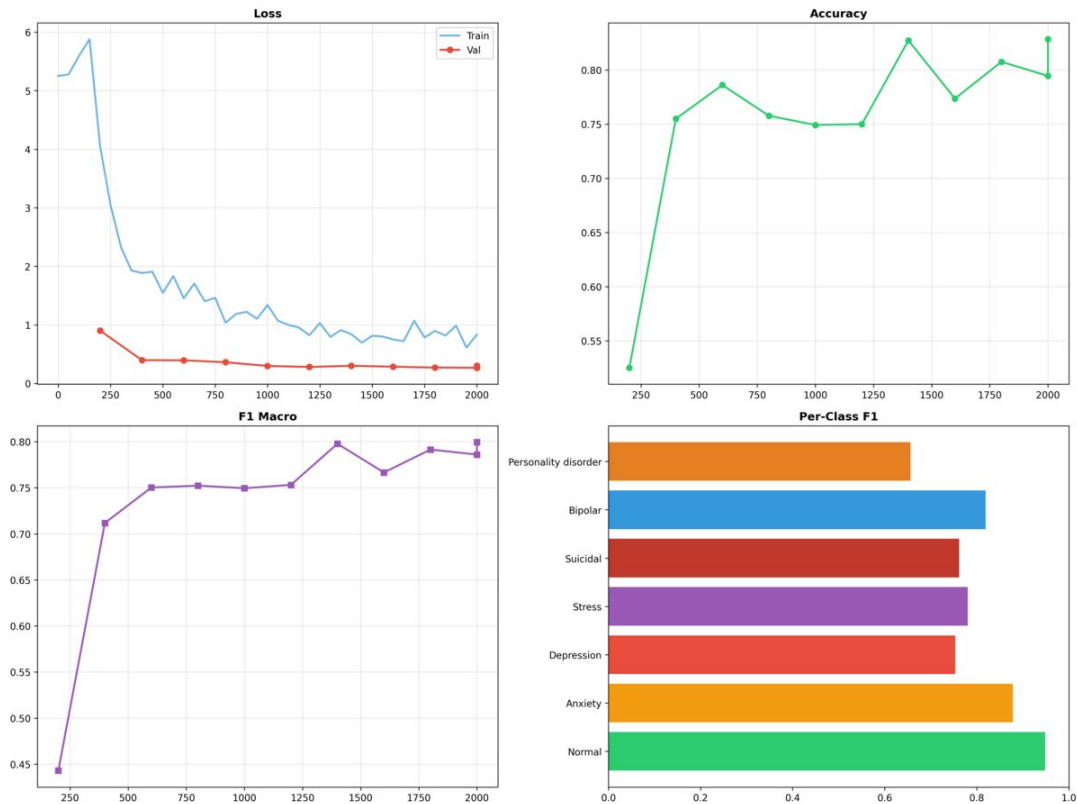


Fig. 5. Model 2 Combined Training Curves: Loss, Accuracy, F1 Macro, and Per-Class F1

C. Model 3: RoBERTa-Large with LoRA

The LoRA model has the most volatile early training among all configurations. The initial loss is quite high (around 16) because the randomly initialized LoRA matrices need time to align with the pretrained weights, but convergence is rapid—the loss stabilizes within the first 1000 steps. Accuracy reaches about 81%, and the per-class F1 distribution reveals particular weaknesses in the Stress (0.686) and Personality disorder (0.644) categories. This suggests that with only ~ 0.8 M trainable parameters, the model simply does not have enough capacity to learn discriminative features for the most challenging classes, even though the underlying RoBERTa-Large representations are richer [2].

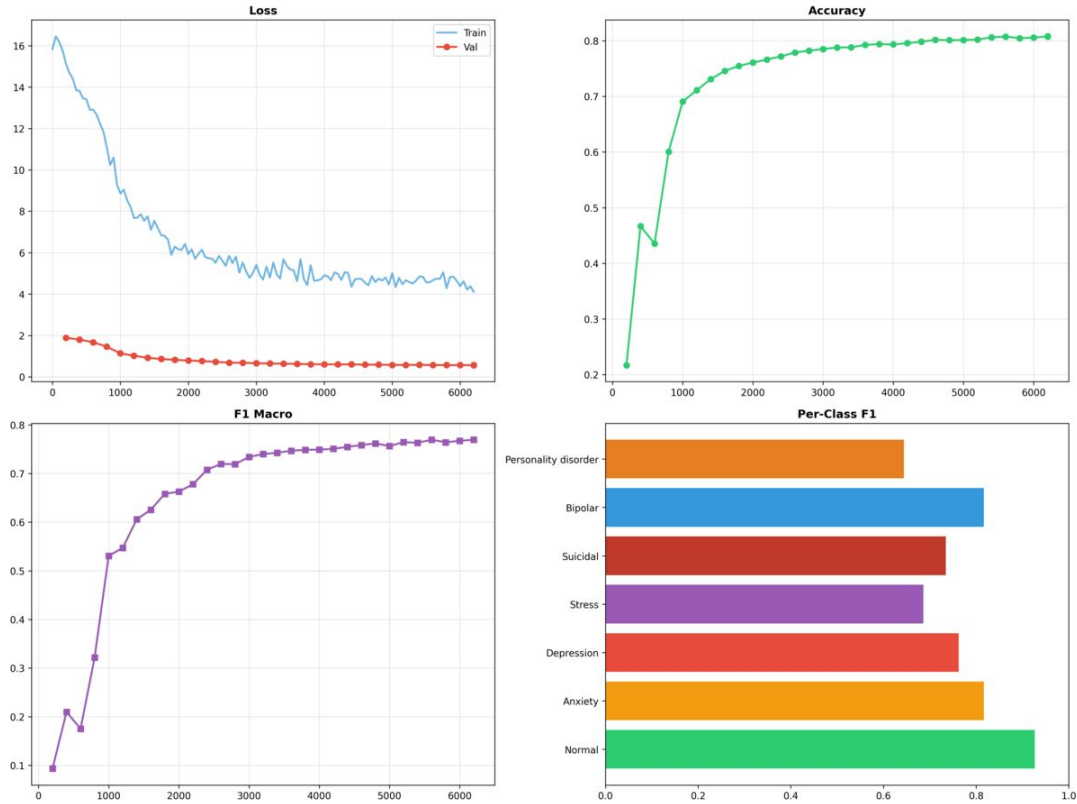


Fig. 6. Model 3 Combined Training Curves: Loss, Accuracy, F1 Macro, and Per-Class F1

D. Model 4: RoBERTa-Base + BiLSTM-CNN Hybrid (Best Model)

The hybrid model shows the smoothest training dynamics of the bunch, thanks to the cosine learning rate schedule. Loss decreases steadily without the spikes seen in other configurations, and accuracy climbs to 85.52%—the highest of all models. The macro F1 of 0.841 is also the best, with notable improvements in Depression (0.802), Bipolar (0.892), and Personality disorder (0.755). What stands out in the per-class F1 bar graph is how much more balanced the performance is across categories compared to the other models: the hybrid architecture does not sacrifice minority-class performance for majority-class accuracy [3].

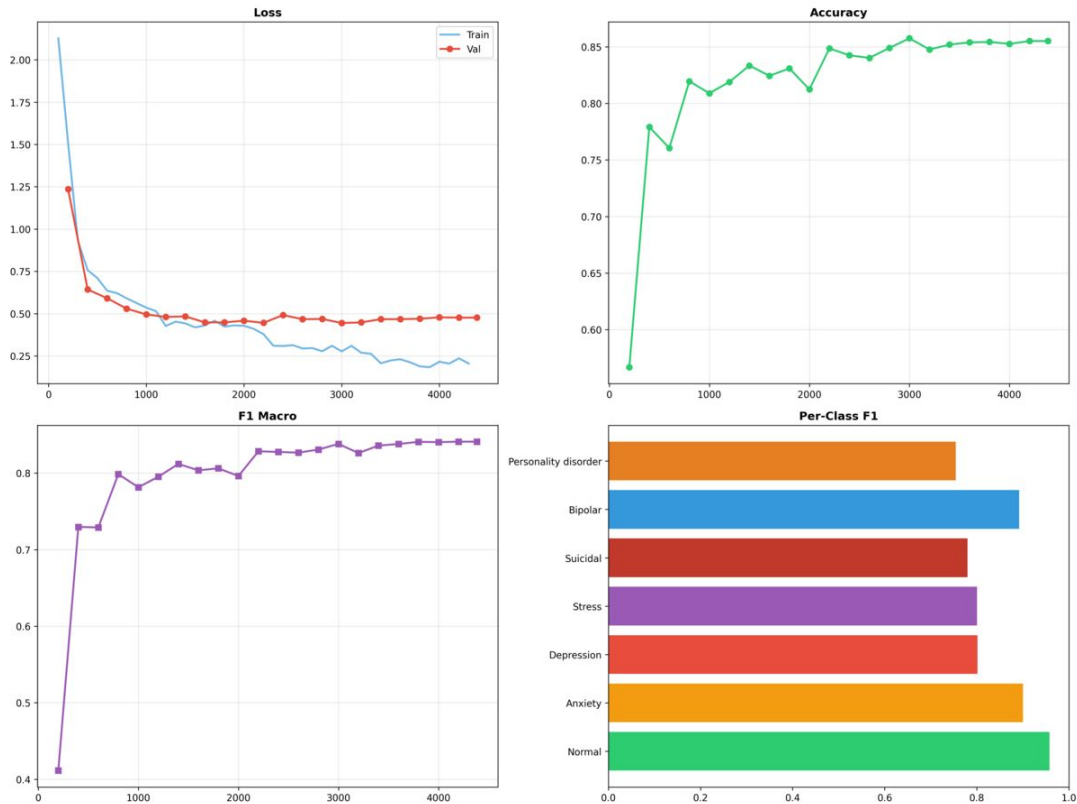


Fig. 7. Model 4 Combined Training Curves: Loss, Accuracy, F1 Macro, and Per-Class F1

E. Chatbot Application Demo

Beyond the training metrics, we also validated the hybrid model through its integration in the Streamlit-based Mental Health Assistant chatbot. Fig. 8 shows a live session where a user describing anxiety-related symptoms receives real-time classification (Anxiety, confidence 0.99) alongside an empathetic conversational response from the Gemini API. The sidebar displays the cumulative Mental Health Score (0.34), the model's prediction, suicidal probability (0.00), and the LLM's independent assessment. The system correctly identifies the user's distress level and responds with appropriate supportive messaging, demonstrating that the trained model generalizes effectively to interactive, real-time use cases.

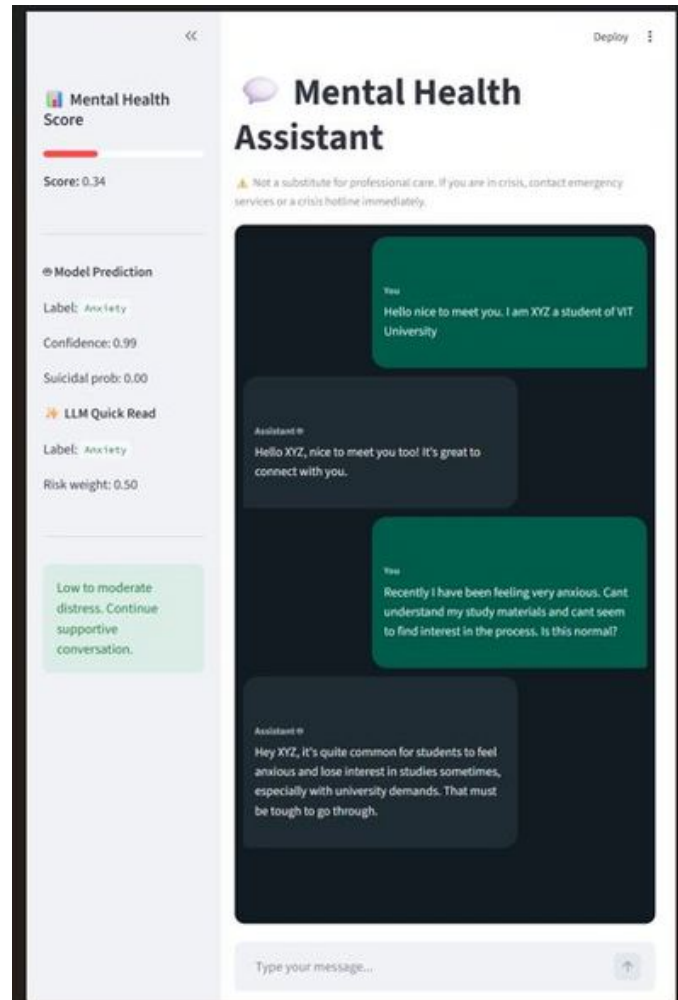


Fig. 8. Mental Health Assistant Chatbot: Live Session with Real-Time Classification and Risk Scoring

IX. DISCUSSION

The results from this study raise several points worth discussing in more depth. This section unpacks the key takeaways, considers the practical implications, and identifies the most important limitations.

A. Why the Hybrid Architecture Works

The hybrid model’s consistent advantage across nearly all classes suggests that mental health text really does contain useful information at multiple linguistic levels. The transformer’s self-attention is great at capturing global document-level semantics—the overall tone and content of a post. But local phrase-level features matter too: negation constructions, intensifiers, and disorder-specific collocations are inherently short-range patterns that benefit from convolutional extraction [3]. And the BiLSTM captures something else entirely: the temporal flow of a narrative, like the progression from describing a triggering event to expressing emotional consequences. By fusing all three perspectives, the hybrid model gets a more complete picture of each post.

The biggest gains appear in intermediate-frequency classes like Depression (+4.6% F1 over the baseline) and Bipolar (+1.3%), where the extra feature diversity helps distinguish between semantically similar but clinically distinct conditions. Even the rare Personality disorder class benefits (+0.9% F1), suggesting that multi-perspective features provide a richer signal for learning with limited examples [14].

B. Domain Adaptation Beats Model Scale

One of the more surprising results is that MentalRoBERTa-base (125M parameters, fully fine-tuned) substantially outperforms RoBERTa-Large (355M parameters) with LoRA. The practical takeaway is clear: for mental health NLP, spending your compute budget on domain-specific pretraining pays off more than throwing a bigger model at the problem, at least when fine-tuning capacity is constrained. MentalRoBERTa has already internalized the vocabulary and discourse patterns of mental health text [11], giving it a much better starting point for classification than a larger but domain-agnostic model.

That said, this comparison is not entirely apples-to-apples. The LoRA model had to use a shorter maximum sequence length due to GPU memory limits, which may have disproportionately hurt performance on longer, more informative posts. Future work could explore LoRA with higher ranks, more target modules, or gradient checkpointing to give the parameter-efficient approach a fairer shot [2].

C. The Focal Loss Paradox

The fact that Focal Loss actually performs worse than standard weighted cross-entropy deserves careful thought. Focal Loss was designed for object detection, where the foreground-to-background ratio can exceed 1:1000. Our class imbalance is much more moderate (roughly 17:1 at its most extreme), so the inverse-frequency weighting in standard cross-entropy may already be sufficient. On top of that, the focusing mechanism adds a new hyperparameter (γ) and changes the gradient landscape in ways that may interact poorly with transformer training dynamics—the attention mechanism already performs a form of implicit difficulty weighting by attending to informative tokens [4][12][15].

D. Clinical Relevance and Ethical Concerns

While these results are encouraging, it is important to be clear about what this system can and cannot do. The dataset labels are based on self-reported or community-assigned categories from social media, not on clinical diagnoses from licensed professionals. This model is best understood as a screening tool—something that can flag potentially at-risk individuals for further evaluation, not a substitute for a proper clinical assessment [13][19].

The ethical stakes are especially high for the Suicidal ideation category. Our best model achieves a per-class F1 of 0.780, which means roughly 22% of suicidal posts could be misclassified. In a real deployment, a false negative—missing a genuinely suicidal individual—could have catastrophic consequences. Any practical system would need robust human oversight, clear escalation protocols, and transparent communication about its limitations. Privacy protections, informed consent, and careful attention to the potential for stigmatization are also essential [17][18].

E. Comparative Perspective with Existing Literature

It is worth situating our results within the broader landscape of mental health text classification. The MIROBERTa paper [1] achieved a best F1 of 0.851 on an eight-class task using an ensemble of domain-adapted and standard RoBERTa models. Our hybrid model reaches 0.841 macro F1 on a seven-class task with a single model (no ensembling), which is competitive given the differences in taxonomy and dataset. Importantly, our hybrid architecture is considerably simpler to deploy than an ensemble, requiring only a single forward pass at inference time. The hybrid CNN-LSTM architecture proposed by Semary et al. [3] achieved 96.28% accuracy on IMDB sentiment analysis, but sentiment classification is a substantially simpler task than multi-class mental health classification, where the categories are more numerous and semantically closer to each other.

The systematic review by Hasan et al. [9] found that transformer-based models generally outperform traditional ML baselines for mental health detection, which is consistent with our findings. However, their review also noted that many studies in this space evaluate on only two or three categories (e.g., depressed vs. not depressed), whereas our seven-class setup more closely mirrors the complexity of real-world clinical screening. The relatively modest gap between our domain-adapted baseline (Model 1, macro F1 0.832) and the hybrid model (Model 4, macro F1 0.841) suggests that while architectural innovations do help, the quality of the pretrained representations remains the single most important factor in this task [11][16].

F. Limitations

This study has several limitations that should temper the conclusions. The dataset is exclusively English-language social media, limiting generalizability to other languages, cultural contexts, and communication modalities such as clinical interviews or chat-based therapy platforms. The seven-class taxonomy does not handle comorbidity—people often experience multiple conditions simultaneously, which a single-label framework cannot capture. There is no temporal modeling: we classify posts in isolation, without tracking how someone’s language changes over time. The absence of clinical ground truth means that label noise is inevitable, as self-reported and community-assigned categories may not align with formal diagnostic criteria. And finally, the computational demands of training and evaluating transformer models may be a barrier for some research groups, particularly in lower-resource settings [9][16].

X. CONCLUSION

This project set out to compare four different strategies for classifying social-media text into mental health categories, and the results tell a clear story. The hybrid RoBERTa-BiLSTM-CNN model is the winner, achieving the highest accuracy (85.52%), macro F1 (0.841), and weighted F1 (0.856) across all configurations. This validates the central idea of the project: that combining global, sequential, and local feature extraction into a single pipeline produces more informative text representations than any single paradigm [3].

Domain-adaptive pretraining proves its worth as a consistently strong baseline strategy. MentalRoBERTa (Model 1) achieves 84.15% accuracy without any architectural innovation, confirming earlier findings that in-domain language modeling makes a real difference for mental health NLP [1][11]. Focal Loss, on the other hand, turns out to be a step in the wrong direction for this particular setting, reducing performance relative to standard weighted cross-entropy [15]. And LoRA, while computationally attractive, does not give the model enough room to adapt, landing at the bottom of the performance table [2].

The per-class analysis reveals that the hybrid model’s improvements are not confined to easy or well-represented categories. It achieves the best F1-scores across virtually every class, with especially notable gains in Depression (0.802 vs. 0.756 for the baseline), Bipolar (0.892 vs. 0.879), and the extremely rare Personality disorder class (0.755 vs. 0.746). This balanced performance profile is precisely what is needed for a practical screening tool, where missing a case of suicidal ideation or misclassifying a bipolar episode carries real-world consequences far beyond a few percentage points on a leaderboard [14].

A. Future Work

Several promising directions emerge from this work. First, **cross-lingual and multilingual extensions** would significantly broaden the applicability of mental health NLP. Mental illness is a global phenomenon, but the vast majority of research—including this project—is conducted exclusively on English text. Multilingual transformer models such as XLM-RoBERTa could be adapted using the same domain-adaptive and hybrid strategies explored here, potentially enabling mental health screening in languages where annotated datasets are scarce or nonexistent [9].

Second, **multi-label classification** would better reflect clinical reality, where comorbid conditions are the norm rather than the exception. An individual experiencing both anxiety and depression should not be forced into a single category. Multi-label frameworks with appropriate loss functions (e.g., binary cross-entropy per class) and evaluation metrics (e.g., subset accuracy, micro/macro F1) would provide a more realistic modeling of mental health phenotypes [12].

Third, **temporal modeling** represents an important frontier. Our current approach classifies individual posts in isolation, but mental health is inherently a longitudinal phenomenon. Tracking how an individual’s language evolves over weeks or months—detecting escalation patterns, response to events, or gradual deterioration—could provide far more actionable information than a single-snapshot classification. Recurrent architectures operating over sequences of posts, or attention mechanisms applied across temporal windows, are natural extensions [14][18].

Fourth, **explainability and interpretability** should be prioritized in any clinical deployment. Techniques such as SHAP analysis (as demonstrated by Sao and Lim [1]), attention weight visualization, and gradient-based attribution methods can help clinicians understand why a model flags a particular post, building trust and enabling informed decision-making. Without interpretability, these systems risk being treated as opaque black boxes, which is especially problematic in a domain where the consequences of errors are severe [19].

Finally, **integration with clinical validation data** is essential for moving from research to practice. The current dataset relies on self-reported or community-assigned labels, which may not align with formal diagnostic criteria. Partnering with mental health professionals to create clinically validated evaluation sets, or testing these models in real-world screening workflows alongside trained clinicians, would provide the evidence base needed for responsible deployment. Any such deployment must be accompanied by robust ethical safeguards—privacy protections, informed consent, human oversight, and transparent communication about what these systems can and cannot do [13][17].

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